Planning as a Cognitive Cost that Discourages Goal Switching

Luke Cho[1], J. Branson Byers[2], Yael Niv[1,2]

1. Princeton Neuroscience Institute, Princeton University
2. Department of Psychology, Princeton University

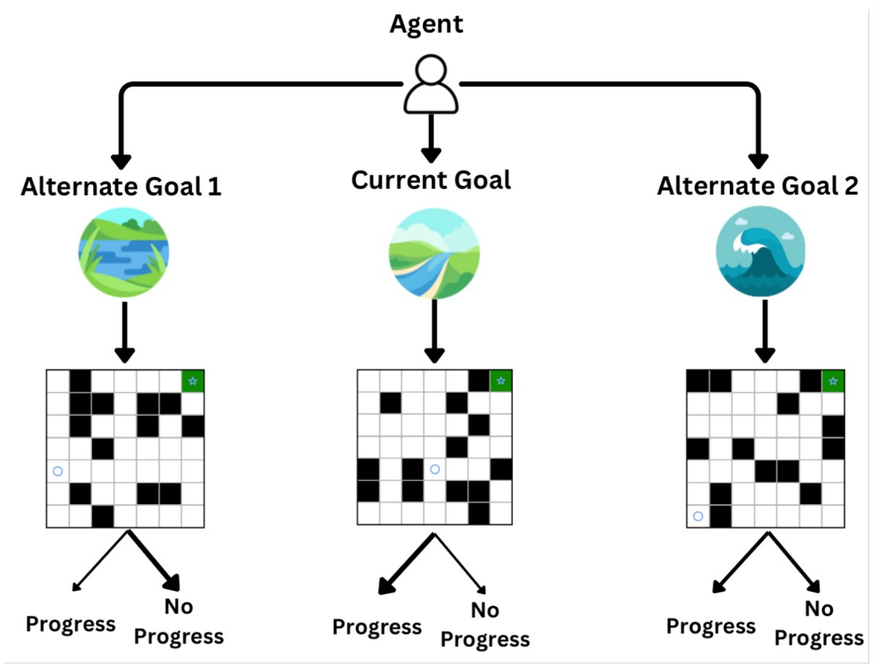
Introduction

- Goals are essential to guiding and sustaining human behavior[1]
- People show "stickiness" to active but suboptimal goals[1]
- Planning towards goals is cognitively costly, and people may be averse to these costs[2]

Does having to re-plan towards a new goal would exert a "planning cost" that discourages individuals from switching their goal?

Task

1. Choose to continue to abandon goal
2. Navigate GridWorld Maze
3. Receive Feedback



- **Non-planning Condition:** participants shown optimal route (plan) outlined in blue, and unable to deviate from it
- **Planning Condition:** participants not shown optimal route; have to create their own plans

Reinforcement Learning Models of Goal Pursuit

- Rescorla-Wagner: $Q_t(g_i) = v_t(g_i) - c_t(g_i)$
- Momentum[3]: $Q_t(g_i) = m_t(g_i)v_t(g_i) - c_t(g_i)$
- Hybrid: $\big(1 - \omega\big(1 - m_t(g_i)\big)\big)\, v_t(g_i) - c_t(g_i)$
  $= \big(\omega \cdot m_t(g_i) + (1 - \omega)\big)v_t(g_i) - c_t(g_i)$

Where:

$$v_t(g_t) = v_{t-1}(g_t) + \alpha\big(r_t - v_{t-1}(g_t)\big)$$

$$m_t(g_i) = \frac{\text{Progress Made on Goal i}}{\text{Size of Goal i}}$$

$$c_t = \begin{cases} 0, & g_i = g_{t-1} \\ \kappa, & g_i \neq g_{t-1} \end{cases}$$

c_t captures goal persistence above and beyond that resulting from goal proximity (m_t) and learned progress probability (v_t)

Results

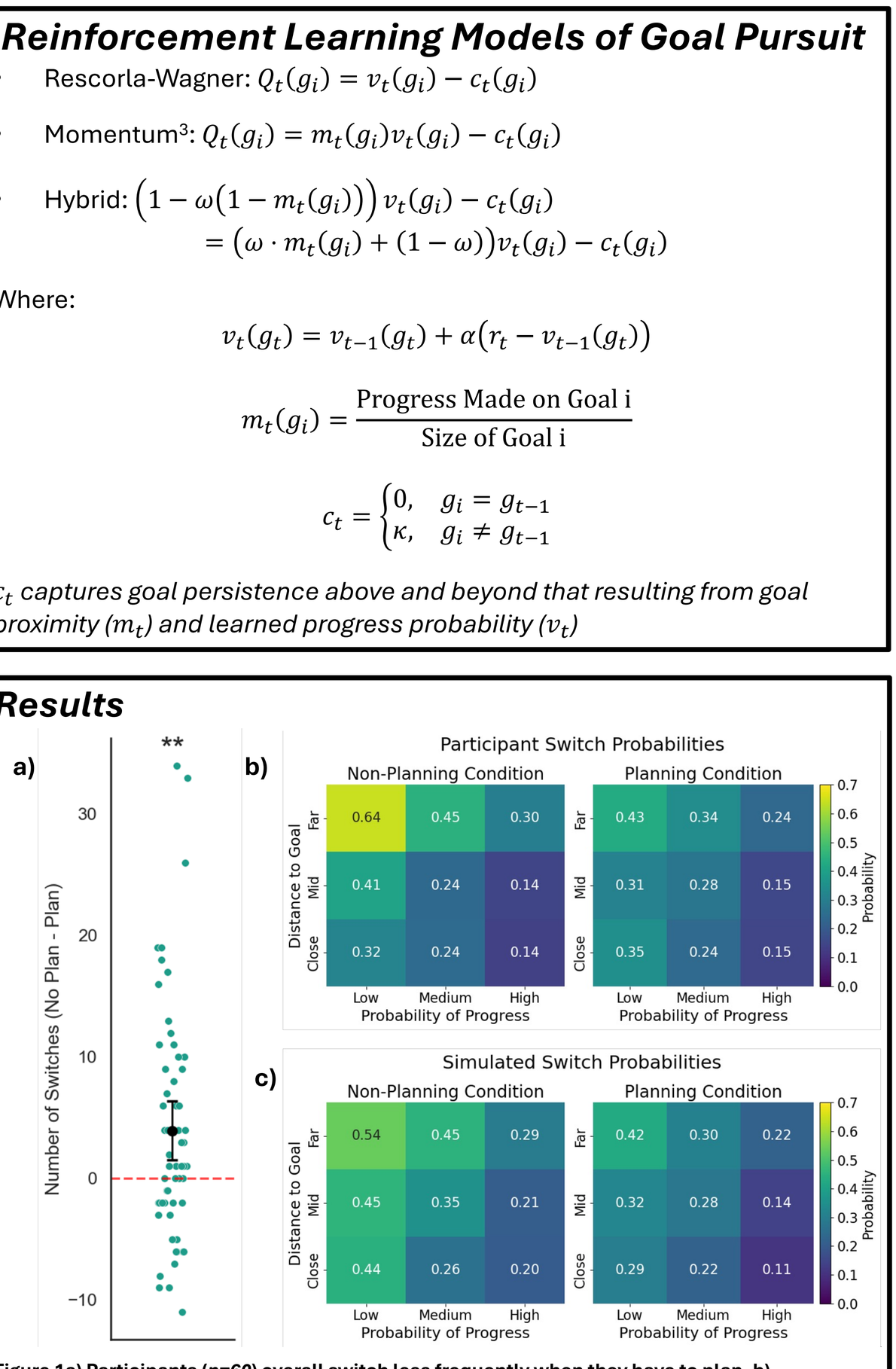


Figure 1a) Participants (*n=60*) overall switch less frequently when they have to plan. b) Participants switch less often holding goal progress probability and accrued progress constant. c) Simulated goal-switch probabilities using hybrid model capture patterns in participant data

Results (cont.)

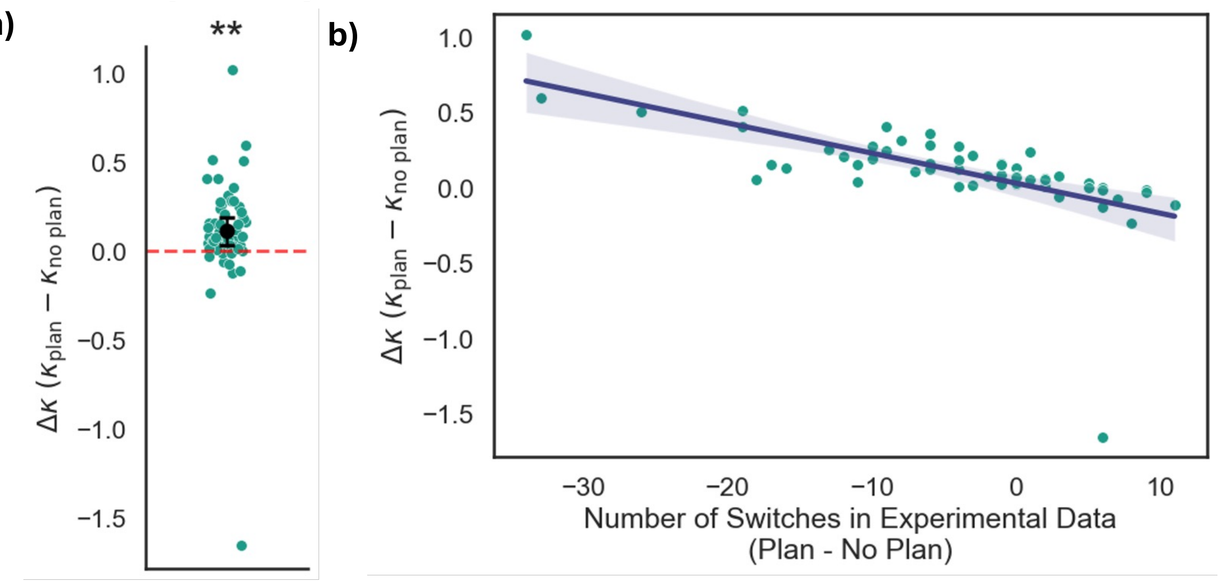


Figure 2a) Choice kernel values using winning hybrid model are higher when individuals have to plan. b) Planning costs (difference between the two choice kernel values) are negatively correlated with the frequency of switching

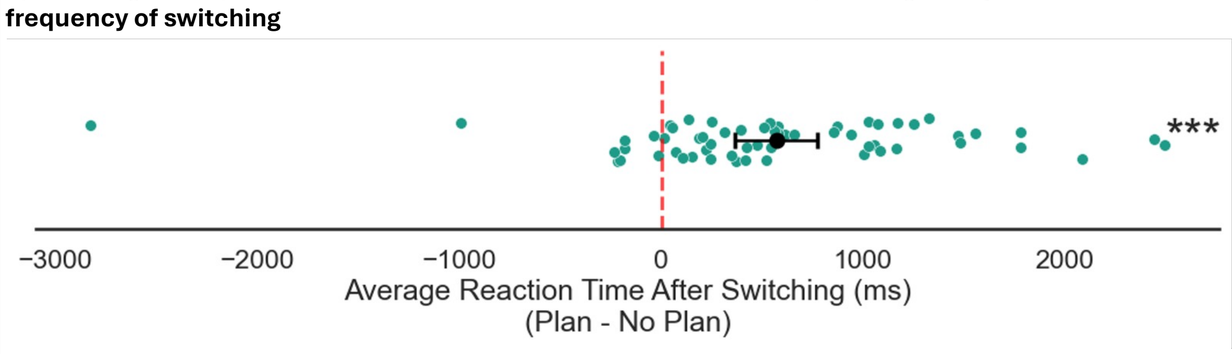


Figure 3) Reaction times to choose navigation direction (step 2) after switching are on average greater when participants have to plan

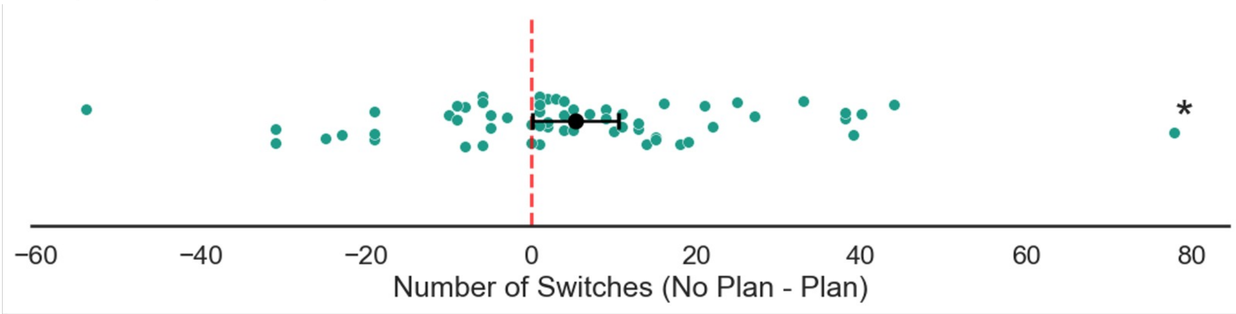


Figure 4) When controlling for the possibility of participants retrieving previous plans from memory, participants still switch less frequently when they plan

Discussion

- We conducted a behavioral experiment that demonstrated people are more likely to abandon their goal in situations where planning is not necessary (plans are provided for them), in contrast to situations in which they have to form their own plan
- Within-participant computational modeling captured goal persistence biases by treating planning costs as a value-based penalty, and demonstrated that this cost was higher when participants had to form their own plans
- Next step: Vary the complexity of these mazes to see if increased difficulty in creating plans will result in greater goal persistence and greater cost kernel values

Acknowledgments

This research was supported by Princeton University's Office of Undergraduate Research Student Initiated Internship Program (OURSIP) through the Hewlett Foundation.

Contact Us!

Extended Abstract QR Code:

Poster PDF QR Code:

Author Emails:

Luke Cho: luke.cho@princeton.edu

Branson Byers: jbbyers@princeton.edu

Citations

1. Holton, E., Grohn, J., Ward, H., Manohar, S. G., O'Reilly, J. X., & Kolling, N. (2024). Goal commitment is supported by vmPFC through selective attention.
2. Callaway, F., van Opheusden, B., Gul, S., Das, P., Krueger, P. M., Griffiths, T. L., & Lieder, F. (2022). Rational use of cognitive resources in human planning.
3. Aenugu, S., & O'Doherty, J. P. (2025). Building momentum: A computational account of persistence toward long-term goals.